ENG. Mech. Eng. 2026
DOI 10.1007/s11465-000-0000-0

RESEARCH ARTICLE

Systematic Design and Locomotion Control of a Modular Bipedal Robot with Closed-chain Leg Mechanism

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ABSTRACT To overcome the challenges of structural compliance and excessive torque requirements encountered by bipedal robots during highly dynamic locomotion, this paper introduces a novel 8-bar closed-chain knee mechanism derived from Stephenson III topology reconstruction. A recursive Newton-Euler dynamic formulation is developed to accurately characterize the system dynamics, while a TOPSIS-based multi-objective optimization framework is employed to simultaneously minimize peak actuation torque, enhance torque smoothness, and preserve the desired workspace characteristics. Building upon the optimized mechanical design, a PPO reinforcement learning controller is trained in Isaac Lab using massively parallel simulations to achieve robust and stable gait generation. Experimental results demonstrate that the proposed optimization strategy maintains a singularity-free global workspace while reducing peak driving torque by 22.1% and improving torque smoothness by 90.9%. Furthermore, validation on a physical prototype confirms the reliability of the Sim-to-Real transfer process, enabling stable walking performance with Roll and Pitch attitude fluctuations constrained within ±2.5° and ±4.0°, respectively, closely matching the stability characteristics of human locomotion. Overall, the proposed integrated mechanism-design and control framework significantly improves locomotion efficiency and establishes a comprehensive methodology for closed-chain bipedal robotic systems, encompassing mechanism optimization, dynamic control, and reliable Sim-to-Real policy deployment.

KEYWORDS Single-leg closed-chain mechanism, Dynamic optimization, Bipedal robot, Reinforcement learning, Newton-Euler formulation, Sim-to-Real transfer.

1 Introduction

To improve the adaptability and payload capacity of bipedal robots operating in complex and unstructured environments, extensive research has been devoted to the development of advanced leg architectures and control strategies [1-4]. Conventional bipedal robotic systems predominantly employ serial open-chain configurations with gait planning methodologies tailored to their kinematic structure [5-7]. These mechanisms are widely favored due to their large reachable workspace and relatively simple kinematic modelling. Nevertheless, under high-load or highly dynamic operating conditions, serial structures are susceptible to elastic deformation and error accumulation, resulting in increased demands on joint actuation torque and reduced overall mechanical efficiency [8-9]. In contrast, closed-chain mechanisms offer superior structural rigidity and transmission efficiency through multi-link coupling and load-sharing characteristics. Owing to these intrinsic mechanical advantages, closed-chain

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architectures have attracted significant attention in the development of high-performance legged robotic systems [10-13].

The structural morphology of a closed-chain mechanism plays a decisive role in determining the locomotion capability and dynamic performance of bipedal robots. To enhance load-bearing capacity and motion stability, researchers have proposed a variety of closed-chain leg configurations based on parallelogram structures and multi-link, multi-loop coupling mechanisms [14-16]. Through inherent kinematic constraints, these robots achieve improved terrain adaptability and robust support for heavy payloads [17, 18]. However, the presence of multiple coupled loops introduces highly nonlinear kinematic and dynamic behaviors, substantially increasing the complexity of mechanism analysis and design. Traditional design methodologies often rely heavily on heuristic parameter tuning and designer experience, limiting the ability to fully exploit the performance potential of sophisticated closed-chain architectures [19-21]. Furthermore, optimizing such hybrid leg mechanisms with respect to actuation torque, energy efficiency, and dynamic performance remains a significant challenge. Precise modelling of complex internal joint forces, coupled dynamic interactions, and globally optimized parameter configurations is essential for achieving smooth, efficient, and stable locomotion, yet these requirements considerably complicate the overall design and optimization process [22-25].

Traditional bipedal gait planning methods predominantly rely on Zero Moment Point (ZMP) theory and simplified inverted pendulum models to maintain dynamic balance and ensure locomotion stability [26-28]. While these approaches have demonstrated considerable success in conventional serial robotic systems, they have become increasingly inadequate for closed-chain leg mechanisms characterized by strong kinematic coupling, multiple loop constraints, and highly linear dynamic interactions. In such systems, deriving accurate analytical dynamic models for rapid gait transitions and posture regulation is often mathematically intractable [29, 30]. As a result, conventional model-based approaches struggle to efficiently generate agile and adaptive locomotion behaviors for complex closed-chain robotic platforms. To overcome these limitations and more fully exploit the intrinsic capabilities of advanced robotic hardware, recent research has progressively shifted toward model-free and data-driven motion planning paradigms [31, 32].

Driven by advances in Reinforcement Learning (RL), learning-based control has emerged as a powerful framework for robotic locomotion. In particular, model-free RL has been widely adopted for the development of autonomous locomotion controllers [33, 34]. By optimizing task-oriented reward functions and leveraging massively parallel training environments, RL enables robots to iteratively refine control policies and acquire sophisticated motor skills through interaction with the environment [35, 36]. Compared with traditional analytical and heuristic control strategies, RL exhibits superior adaptability to complex nonlinear dynamics and diverse operational scenarios, while reducing dependence on manually engineered control rules. Consequently, RL has achieved remarkable success in the locomotion control of high DOF open-chain legged robots [37-39]. Despite these advances, extending RL-based control frameworks to multi-loop closed-chain parallel robots remains a formidable challenge. The presence of strict geometric constraints, coupled dynamics, and physically constrained motion spaces significantly increases the complexity of policy learning and stability optimization. Moreover, achieving reliable transfer of learned policies across simulation platforms and from simulation to physical hardware (Sim-to-Real transfer) continues to be a critical obstacle for the deployment of closed-chain robotic systems in real-world environments [40, 41].

The primary contributions of this work are summarized in a unified framework that systematically integrates mechanism optimization with learning-based control for closed-chain bipedal robots. First, a Stephenson-III-derived 8-bar leg mechanism with a singularity-free workspace is proposed. A global Newton-Euler formulation resolves the underdetermined internal forces, enabling an entropy-weighted TOPSIS optimization that reduces peak torque by 22.1% and improves torque smoothness by 90.9%. A model-free reinforcement learning controller, trained in Isaac Lab and validated via Sim-to-Sim cross-engine tests, is deployed on an 8-DOF prototype, achieving stable human-like gaits and robust Sim-to-Real transfer.

The remainder of this paper is organized as follows: Section 2 presents the closed-chain reconfigurable mechanism design and provides a comprehensive analysis of the robot kinematics and workspace characteristics. Section 3 develops the dynamic model based on the Newton-Euler formulation and details the proposed multi-objective optimization procedure and corresponding results. Section 4 describes the RL framework, including policy training in simulation, mechanism analysis, and cross-engine validation. Section 5 introduces the design and implementation of the physical robot prototype and experimentally evaluates the feasibility and energy efficiency of the proposed real-world gait data. Finally, Section 6 concludes the paper and discusses future research directions.

2 Integrated Robot Configuration Design and Kinematic Analysis

This section presents the systematic design of the proposed bipedal robot architecture together with its kinematic characterization. To satisfy the demanding requirements of high payload capacity, structural rigidity, and dynamic locomotion stability, a novel 8-bar closed-chain leg mechanism is first synthesized through topological reconstruction. Subsequently, a comprehensive kinematic framework is established to investigate the foot trajectory and systematically analyze the singularity-free global workspace of the mechanism. These analyses provide the structural and theoretical foundation for the dynamic modeling, optimization, and reinforcement learning control strategies introduced in the subsequent sections.

2.1 Single-Leg Closed-Chain Mechanism Design

To meet the requirements of high load-bearing capability and enhanced structural stiffness of bipedal robots during dynamic locomotion, this paper proposes a novel 8-bar coupled lower-limb mechanism inspired by the motion characteristics of spatial single-loop systems capable of achieving stable and continuous cyclic locomotion.

The single-leg design of this robot adopts a modular hybrid serial-parallel architecture incorporating four actively actuated degrees of freedom (DoFs). From proximal to distal, these include: the Hip_roll, responsible for maintaining lateral body balance; the Hip_pitch, used to regulate stride generation and forward locomotion; the Ankle Pitch, enabling terrain adaptability and shock absorption; and the core Knee_pitch, serving as the primary power-generation of the lower limb for large-scale flexion and extension motions.

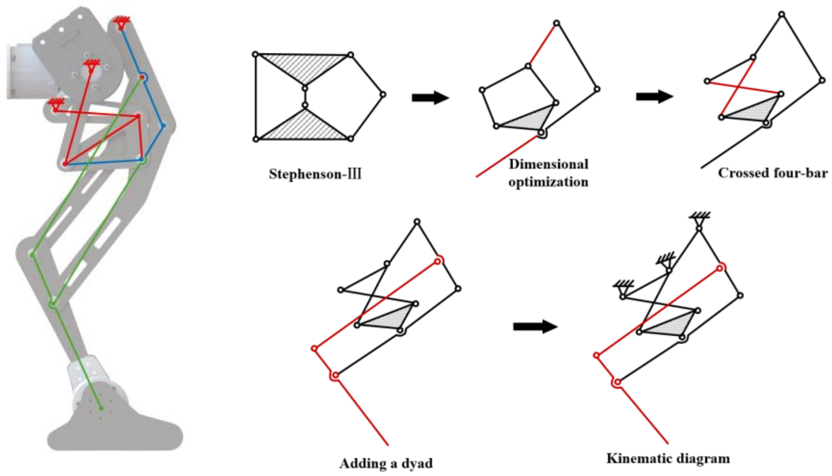


Fig. 1. Topological evolution of the 8-bar closed-chain knee mechanism: From Stephenson-III to an 8-bar three-loop mechanism.

To validate the independent actuation requirements of the proposed 8-bar closed-chain reconfigurable mechanism under a general configuration, its DOF is evaluated using the classical Grübler-Kutzbach (G-K) criterion:

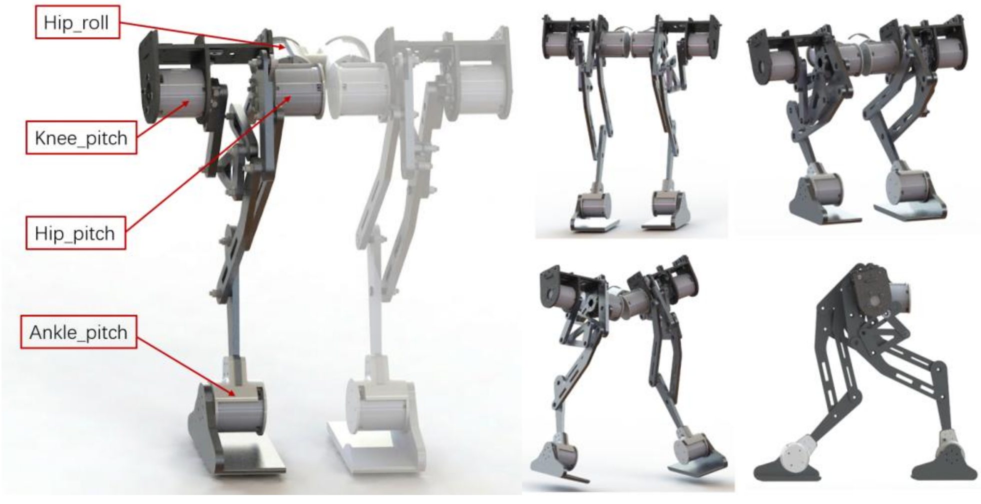
$$M = d\left(n - g - 1\right) + \sum_{i=1}^{g} f_i + v \tag{2-1}$$

Where d = 3 denotes the motion-space dimension for the planar mechanism or its equivalent projection. Within the core loop of the knee module, the number of moving links n, the number of revolute joints g, and the local redundant geometric constraints v offset each other, resulting in an effective DOF of M = 1 for its local topology. This analysis indicates that only a single actively driven joint motor, integrated with an encoder for servo-state feedback, is required at the proximal driving crank to fully constrain and coordinate the motion of all 8 links along a prescribed coupled trajectory, while the remaining revolute joints operate as passive followers. Such a "single-input, multi-motion" configuration not only ensures precise and independent control of lower-limb locomotion but also significantly reduces structural complexity, actuator mass, and energy consumption.

2.2 Overall Architecture Design and Symmetrical Layout

Building upon the proposed single-leg module, this section presents the complete bipedal robot architecture designed to achieve stable and efficient locomotion. The robot adopts a strictly mirror-symmetrical left-right configuration, which not only simplifies dynamic modeling during gait planning but also promotes balanced lateral force distribution in the static phase.

As illustrated in **Fig. 2(a)**, the robot's mechanical components are integrated within a compact and highly optimized structural framework. To enhance dynamic responsiveness, the system follows a proximal-centered lightweight design philosophy, in which the heavier servo motors are concentrated near the pelvis and thighs (i.e., the Hip_roll, Hip_pitch, and Knee_pitch joints). In contrast, the distal Ankle joint is primarily driven through the proposed high-efficiency closed-chain transmission mechanism described above. This mass distribution strategy produces a lightweight extension toward the distal extremities (with the distal module weighing only 0.4 kg, corresponding to approximately 10.5% of the total leg mass), which significantly reduces the moment of inertia of the leg during high-speed swing phases and improves the servo response bandwidth. Furthermore, as shown in **Fig. 2(b)**, the robot maintains a wide range of motion across extreme joint configurations without experiencing internal mechanical interference. The resulting kinematic envelope establishes clear physical operational boundaries—specifically, a maximum anterior-posterior stride capacity of 679 mm and a lateral balance margin of 401.6 mm. These characteristics provide substantial motion redundancy and operational safety margin, thereby creating a robust physical foundation for subsequent reinforcement learning-based gait generation and dynamic locomotion exploration.



(a) (b)

Fig. 2. Overall architecture: (a) Mechanical morphology, and (b) extreme joint configurations.

This structural configuration adopts a lightweight distal-extension design strategy, in which the distal module weighs only 0.4 kg, representing merely 10.5% of the total leg mass. Such a mass distribution effectively reduces rotational inertia during high-speed swing motions and enhances the servo system's dynamic response bandwidth. Moreover, the proposed architecture establishes well-defined kinematic performance boundaries, including a maximum sagittal stride length of 679 mm and a lateral balance margin of 401.6 mm. These quantified motion capabilities provide substantial kinematic redundancy and fault-tolerance margins, thereby creating a reliable physical foundation for subsequent deep reinforcement learning-based autonomous gait exploration in forward locomotion tasks.

2.3 Closed-Chain Kinematic Modeling and Single-DOF Foot Trajectory

This section develops the kinematic constraint model for the proposed 8-bar closed-chain leg mechanism. Owing to the topological reconstruction of the mechanism, as illustrated in **Fig. 1**, point E in the conceptual topology coincides with the primary driving hinge point A in the realized physical configuration (i.e., A $\equiv$ E). From a kinematic perspective, the mechanism can therefore be interpreted as a system composed of three nested and strongly coupled closed loops, including a distal parallelogram substructure, which tend to achieve complex posture reconfiguration through the unilateral actuation of the crank.

2.3.1 Mechanism Topology and Link Definitions

The input angle of the driving joint *A* is denoted by θ_{AD}, as illustrated in **Fig. 3**, which presents the schematic configurations of each kinematic loop. Based on the geometric constraints of the entire mechanism, the overall system can be decomposed into a two-stage set of nonlinear equations:

1. Stage 1: The Base Four-Bar Loop ADCB

This loop forms the primary transmission structure of the mechanism. Using the Euler vector loop formulation, the closed-loop kinematic constraint for loop *A-D-C-B-A* can be written as:

$$\begin{cases} \mathrm{r_{AD}}\cos\theta_{\mathrm{AD}} + \mathrm{r_{DC}}\cos\theta_{\mathrm{DC}} - \mathrm{r_{AB}}\cos\theta_{\mathrm{AB}} - \mathrm{r_{BC}}\cos\theta_{\mathrm{BC}} = 0 \\ \mathrm{r_{AD}}\sin\theta_{\mathrm{AD}} + \mathrm{r_{DC}}\sin\theta_{\mathrm{DC}} - \mathrm{r_{AB}}\sin\theta_{\mathrm{AB}} - \mathrm{r_{BC}}\sin\theta_{\mathrm{BC}} = 0 \end{cases} \tag{2-2}$$

2. Stage 2: The Five-Bar Coupled Loop EFGHI

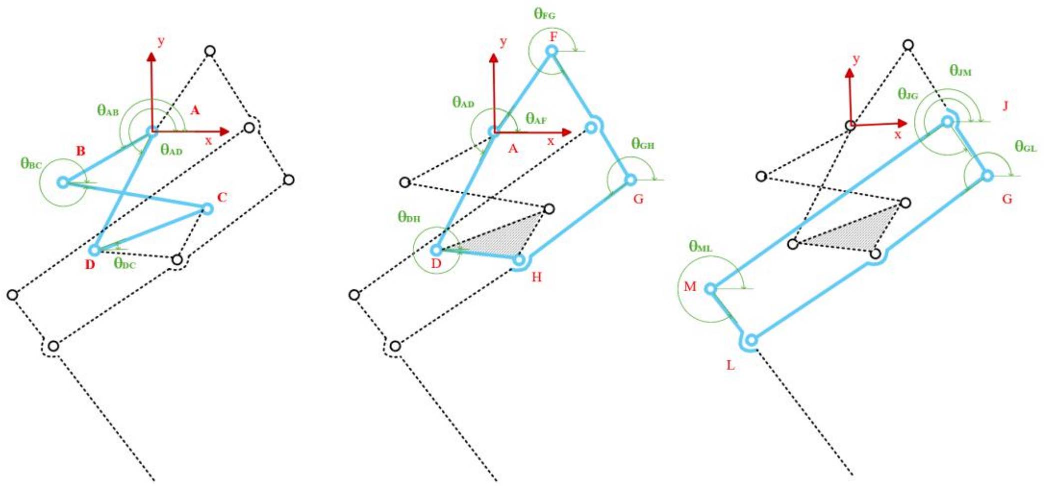
After determining θ_{DC}, the vector loop equation for the secondary loop E-F-G-H-I-E is established according to the kinematic coupling relationships among the connected links (e.g., $\theta_{IH} = \theta_{DC} - \phi_{bias}$):

$$\begin{cases} \mathrm{r_{EF}}\cos\theta_{\mathrm{EF}} + \mathrm{r_{FG}}\cos\theta_{\mathrm{FG}} + \mathrm{r_{GH}}\cos\theta_{\mathrm{GH}} - \mathrm{r_{IH}}\cos\theta_{\mathrm{IH}} - \mathrm{r_{AD}}\cos\theta_{\mathrm{AD}} = 0 \\ \mathrm{r_{EF}}\sin\theta_{\mathrm{EF}} + \mathrm{r_{FG}}\sin\theta_{\mathrm{FG}} + \mathrm{r_{GH}}\sin\theta_{\mathrm{GH}} - \mathrm{r_{IH}}\sin\theta_{\mathrm{IH}} - \mathrm{r_{AD}}\sin\theta_{\mathrm{AD}} = 0 \end{cases} \tag{2-3}$$

3. Stage 3: Geometric Constraints of the Distal Parallelogram Loop JGLM

The distal section of the mechanism adopts a parallelogram closed-chain configuration, which imposes strict geometric constraints. Specifically, the line segment JG on link 4 remains continuously parallel to the line segment ML on link 7 at all times, implying that the orientation angle θ_{ML} of link MLN is always equal to θ_{JG}. Likewise, the pushrod JM is parallel to link GL. The pushrod JM, corresponding to link 6, therefore has its orientation angle θ_{JM} strictly constrained to be equal to θ_{GL}.

By numerically solving the above set of nonlinear equations simultaneously, the instantaneous angular positions of all links in the mechanism can be obtained.



(a) (b) (c)

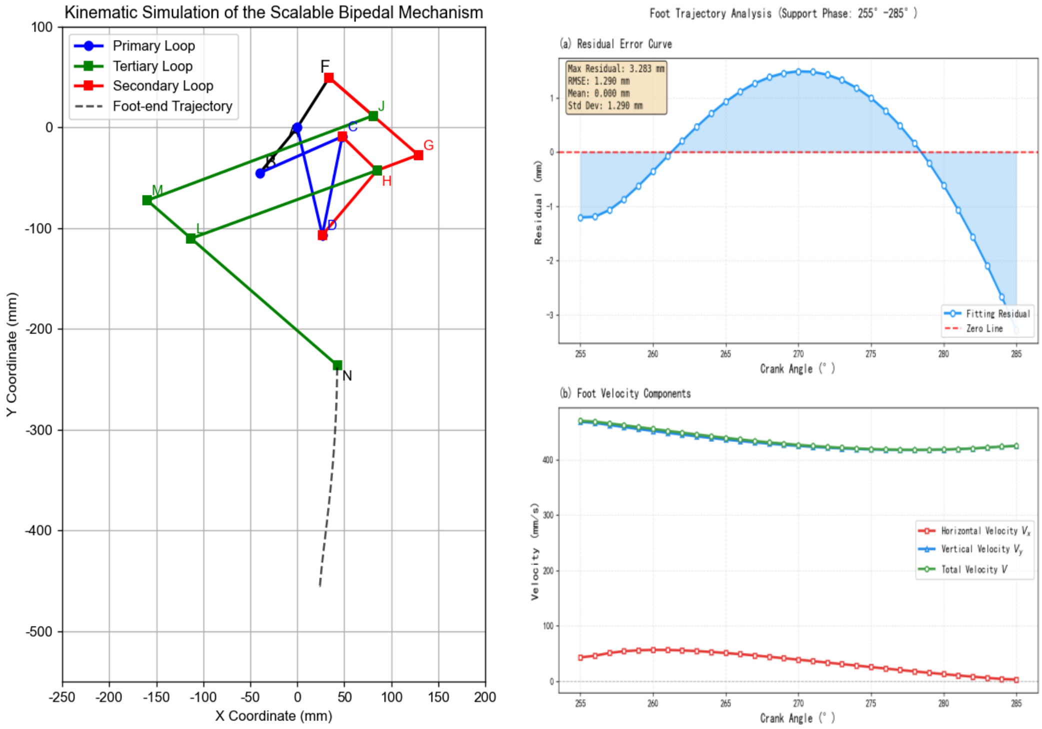
Fig. 3. Schematic diagrams of the three kinematic loops: (a) first loop, (b) second loop, (c) third loop.

2.3.2 Foot Coordinate Mapping and Single-DOF Characteristics

Based on the previously derived angular solutions, the position vector of the foot point N relative to the robot coordinate origin A, denoted as $P_N = [x_N, y_N]^T$, can be obtained through a series of cascaded coordinate transformation matrices:

$$\boldsymbol{P}_N = \boldsymbol{P}_F + \boldsymbol{R}(\theta_{FG})\boldsymbol{L}_{FG} + \boldsymbol{R}(\theta_{GH})\boldsymbol{L}_{GH} + \ldots + \boldsymbol{R}(\theta_{MN})\boldsymbol{L}_{MN} \tag{2-4}$$

Since the mechanism possesses a single degree of freedom (*M=1)*, all kinematic states of the foot, including displacement, velocity, and acceleration, are uniquely determined by the input angle $\theta_{AD.}$ As illustrated in **Fig. 4(a)**, this strong kinematic coupling enables the foot to follow a predefined nonlinear trajectory using only a single servo actuator. To further evaluate the accuracy of the kinematic mapping, **Fig. 4(b)** presents the residual error curves and the foot tip velocity profile during the support phase. The results indicate that the geometric deviations remain within a very small range while the foot tip velocity varies smoothly, confirming minimal impact loading during ground contact. This inherent structural property ensures high-fidelity trajectory tracking and provides a reliable foundation for subsequent locomotion control.



(a) (b)

Fig. 4 Analysis of foot trajectory and tracking precision: (a) Single-DOF foot trajectory in the mechanism coordinate system, and (b) geometric deviation residuals and foot tip velocity distribution.

2.4 Overall Workspace and Reachability Analysis

To comprehensively evaluate the three-dimensional kinematic capabilities of the bipedal robot, this section presents a systematic analysis of the reachable workspace of the entire robot's foot. The analysis is established based on the kinematic model of the previously described single-DOF knee mechanism, integrated with the serial kinematic chain formed by the Hip_roll and Hip_pitch joints.

2.4.1 Coordinate System Definition and Joint Configuration

The robot employs a right-handed coordinate system with its origin located at the center of the body base (Base Link). The *X*-axis points forward, the *Y*-axis points to the robot's right side, and the *Z*-axis points vertically upward. The kinematic chain of each leg consists of the following joints:

1) Hip_roll joint: Rotates about the X-axis within a range of [-20°, 20°], providing lateral swing motion of the leg;
2) Hip_pitch joint: Rotates about the Y-axis within a range of [-30°, 30°], controlling the forward and backward swing of the leg;
3) Knee_pitch mechanism: Actuated by the proposed 8-bar closed-chain mechanism, with a crank angle range of [255°, 285°], enabling large-amplitude extension and flexion movements of the foot.

The left and right legs are arranged symmetrically with respect to the robot's *XZ* plane. To maintain kinematic consistency, the Hip_roll joints on the two sides rotate in opposite directions.

2.4.2 Forward Kinematics Solution Process

The solution for the foot position of the entire robot is implemented using a hierarchical recursive approach:

Step 1: Knee joint mechanism solution

For a given crank angle θ_{knee}, the 8-bar knee mechanism is solved using the staged nonlinear equations established in Section 1.3. This process yields the two-dimensional position of the foot point N in the local *XZ*-plane coordinate system of the mechanism: $P_{N,\text{local}} = [x_N, z_N]^T$

Step 2: Hip_pitch rotation transformation

The planar coordinates of the mechanism are then extended into a three-dimensional representation: $P_{N,3D} = [x_N, 0, z_N]^T$.

Subsequently, a rotation transformation about the Y-axis is applied using the Hip_pitch angle through the rotation matrix $R_y(\theta_{hip\_pitch})$:

$$\boldsymbol{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \tag{2-5}$$

$$\boldsymbol{P}_{\text{foot}} = \boldsymbol{P}_{\text{hip\_roll}} + \boldsymbol{R}_x(\theta_{\text{hip\_roll}})(\boldsymbol{P}_{\text{hip\_pitch}} + \boldsymbol{P}_{\text{hip}} - \boldsymbol{P}_{\text{hip\_roll}}) \tag{2-6}$$

2.4.3 Reachable Workspace and Singularity Analysis

To characterize the kinematic limits of the robot, the forward kinematics solver is employed to evaluate the three-dimensional reachable workspace over the full range of active joint motions. Statistical analysis shows that the workspace of a single leg (taking the right leg as an example) covers 679 mm in the anterior-posterior direction (*X*-axis: [-158.5, 520.5] mm), 401.6 mm in the lateral direction (*Y*-axis: [-123.3, 278.3] mm), and 254.4 mm in the vertical direction (*Z*-axis: [-588.6, -334.2] mm). These results demonstrate that the robot possesses considerable stride capability, sufficient lateral balance margin, and adequate vertical motion range for typical terrain adaptation tasks. In addition, numerical evaluations achieve a 100% convergence rate throughout the continuous joint space, confirming that the proposed closed-chain mechanism operates without kinematic singularities.

From the perspective of bipedal symmetry, the mirror-symmetrical arrangement of the two legs extends the overall lateral workplace to 556.6 mm (Y-axis: [-278.3, 278.3] mm), thereby enhancing lateral stability. As illustrated in **Fig. 5**, both the workplace point-cloud distribution and the corresponding solid models visually confirm that the singularity-free and symmetric workspace prevents internal mechanical interference during highly dynamic motions, including large-stride walking and deep squatting.

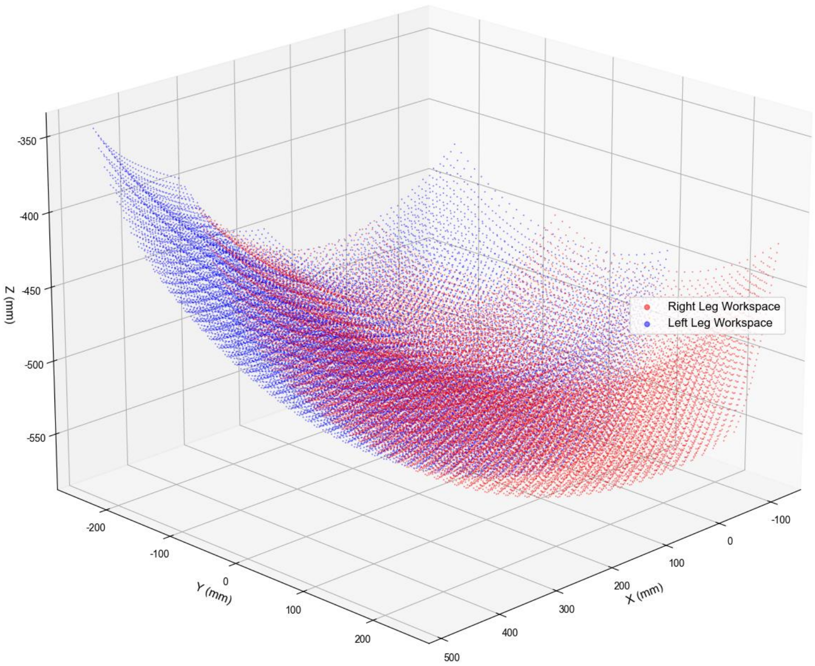


Fig. 5. Bipedal workspace.

2.4.4 Reachability and Gait Planning Constraints

Based on the aforementioned workspace analysis, several important constraints and performance characteristics can be established for subsequent gait planning.

Maximum step-length constraint: The maximum reachable distance of a single leg in the anterior-posterior direction is 679 mm. Considering the alternating support gait pattern in bipedal locomotion, the theoretical maximum stride length can exceed 1.3 m.

Lateral balance margin: The single-leg lateral motion range is approximately 400 mm, enabling effective compensation for lateral disturbances and supporting lateral obstacle-avoidance maneuvers.

Squatting and standing height range: The 254 mm extension-flexion stroke along the Z-axis, together with the body height adjustment, allows the robot to achieve a wide range of postures, from deep squatting to fully upright standing.

Singularity-free operating region: The forward kinematics solution achieves a 100% success rate throughout the entire joint space, indicating the absence of kinematics singularities and ensuring reliable and robust motion control.

Overall, the bipedal robot exhibits a well-structured and sufficiently large reachable workplace capable of satisfying diverse locomotion requirements in a complex environment. This provides a strong kinematic foundation for subsequent dynamic optimization and gait control development.

3 Dynamic Modeling and Multi-Objective Optimization

Building on the kinematic modeling and workspace verification presented in the previous section, this chapter focuses on the dynamic behavior and structural parameter optimization of the bipedal robot. Owing to the multi-loop coupling characteristics of the 8-bar mechanism, the internal force transmission within the system is highly complex. To accurately quantify the required joint torques and internal constraint forces, a comprehensive dynamic model is first established using the Newton-Euler formulation. On this basis, a multi-objective optimization framework is introduced to refine the linkage parameters. The optimization aims to reduce peak actuation torque while improving the overall smoothness of motion. Establishing this optimized structural configuration provides a stronger physical foundation for the mechanism, leading to enhanced energy efficiency and improved hardware reliability during practical locomotion.

3.1 Dynamic Modeling of the Closed-Chain Mechanism Based on the Newton-Euler Formulation

To address the force and torque fluctuations that occur during the motion of this closed-chain parallel mechanism, a global quasi-static dynamic model is established using the Newton-Euler formulation. In terms of physical component modeling, the mechanism primarily consists of two categories of structural components:

Slender Rods: The elongated links within the mechanism (such as r_{AD}, r_{BC}.) are modeled as slender rods with uniformly distributed mass. Their moment of inertia about the center of mass is defined as:

$$I_{\mathrm{rod},i} = \frac{1}{12} m_i L_i^2 \tag{3-1}$$

Triangular Rigid Body Components: Key functional assemblies in the mechanism (such as the triangular linkage group formed by r_{DC}, r_{CH}, and r_{DH}) are modeled as planar rigid bodies. Their moments of inertia are obtained by integrating the mass distribution over the planar area, which is incorporated into the dynamic formulation as:

$$I_{\mathrm{tri}} = \iint \left(x^2 + y^2\right) \rho dA \tag{3-2}$$

For such rigid-body components, the force analysis must simultaneously account for the geometric constraints imposed by the three vertices and the dynamic variation of the center-of-mass position during motion.

For each principal link body in the mechanism, the translational and rotational equilibrium equations are formulated according to d'Alembert's principle as follows:

$$\sum \boldsymbol{F}_{\mathrm{ext}} - m_i \boldsymbol{a}_{G,i} = 0 \tag{3-3a}$$

$$\sum M_{G,i} - I_i \alpha_i = 0 \tag{3-3b}$$

Where $a_{G,i}$ denotes the centroid acceleration of the corresponding component (either a link or a triangular part), and α_i represents its angular acceleration.

Because the closed-chain mechanism contains multiple interconnected kinematic loops, the interaction forces among the joints are strongly coupled. To determine all internal constraint forces together with the required driving torque, the equilibrium equations of the entire mechanism are solved simultaneously, resulting in a global force matrix equation containing 21 unknown variables:

$$\boldsymbol{A}_{21\times 21}(\theta)\cdot \boldsymbol{x} = \boldsymbol{B}(\theta,\omega,\alpha) \tag{3-4}$$

Where the unknown vector $\mathbf{x} \in \mathrm{R}^n$ is defined as:

$$\boldsymbol{x} = \left[F_{A,x}, F_{A,y}, F_{B,x}, F_{B,y}, F_{C,x}, \ldots, F_{N,y}, \tau_{\text{in}} \right]^T \tag{3-5}$$

The vector x includes all hinge point forces from the base fulcrum A to the terminal foot point N, as well as the input driving torque τ_{in} generated by the motor.

The coefficient matrix A characterizes the geometric topology and constraint relationships of the mechanism. For a representative link connecting joints J and K, the corresponding submatrix within A can be expressed as:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ -(y_J - y_G) & (x_J - x_G) & -(y_K - y_G) & (x_K - x_G) & \delta_{\text{in}} \end{bmatrix} \tag{3-6}$$

Where (x_G, y_G) represent the centroid coordinates of the corresponding link, and δ_{in} denotes the discriminant coefficient associated with the input shaft. The right-hand-side vector B includes the gravitational force term $m_i g$, the inertial force term $m_i a_{Gi}$, and the inertial moment term $I_i \alpha_i$. By solving this large-scale system of linear equations in real time, the continuous motor torque profile over the entire operating cycle can be obtained. This cycle corresponds to the crank rotation range of 255° to 285° for the single-DOF foot trajectory. The resulting torque curve provides an accurate physical foundation for the subsequent multi-objective optimization process.

3.2 Dynamic Model Verification and ADAMS Benchmarking

To validate the accuracy of the proposed Newton-Euler dynamic model, the analytical results are benchmarked against numerical simulations performed using the ADAMS multi-body dynamics platform. Under identical mechanism parameters and kinematic input conditions—specifically a driving angle θ_{AD} ranging from 255° to 285° and a constant angular velocity ω = 2 rad/s—the dynamic responses of the system are evaluated using the following two approaches:

1. Solving the driving torque T and the corresponding hinge constraint forces using the 21×21 global matrix equation established in this paper.

2. Implementing the same kinematic constraints in the ADAMS simulation model and extracting the driving-joint reaction torque together with the hinge reaction forces.

As illustrated in **Fig. 6**, the driving torque profiles and the constraint forces at the key hinge joints obtained from the two methods exhibit strong agreement throughout the entire motion cycle. Statistical analysis indicates:

4) Peak driving torque error: < 3.5%
5) Root Mean Square Error (RMSE): 0.042 N·m
6) Correlation coefficient R^2: 0.987
7) Average relative error of key hinge forces at points D, G, and H: < 5%

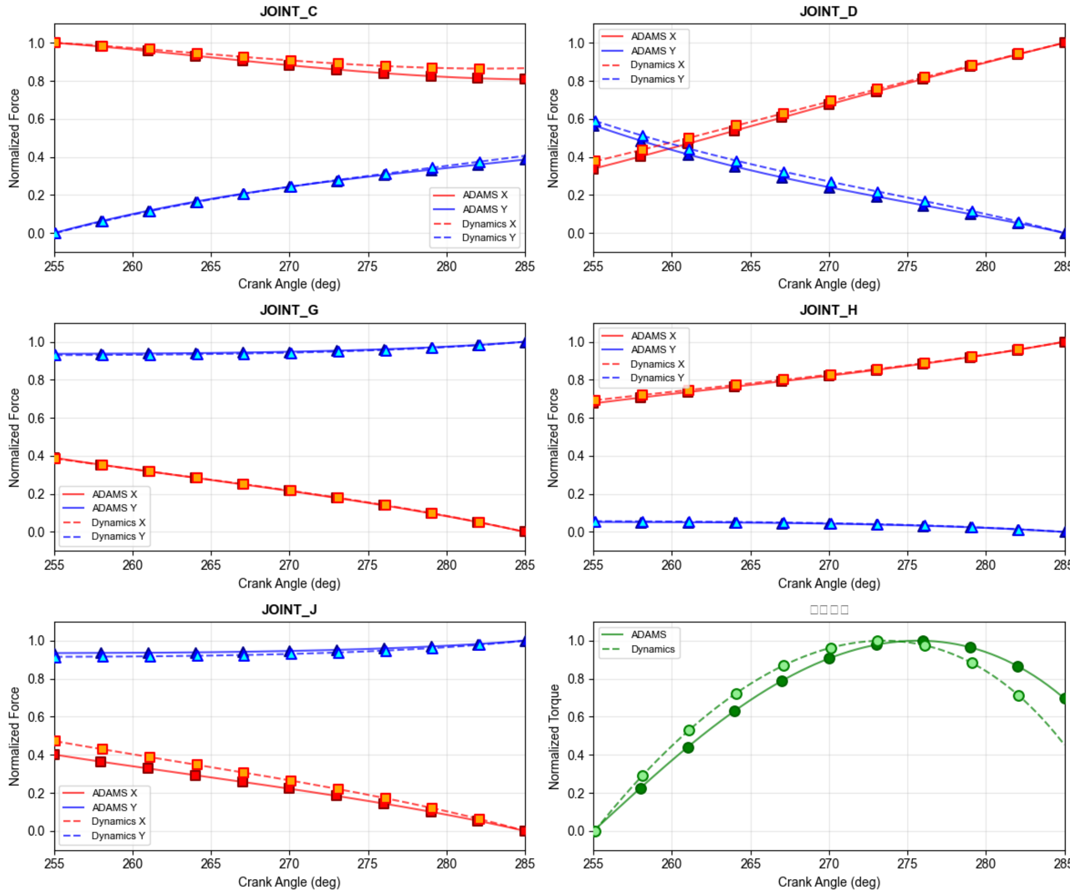


Fig. 6. Comparison between theoretical calculations and simulation results.

This comparison confirms the accuracy of the dynamic model and the numerical stability of the proposed solution method, thereby providing a reliable theoretical foundation for subsequent parameter optimization based on this model.

3.3 Definition of Optimization Objective Functions and Evaluation Metric System

To achieve optimal reconfiguration of the mechanism parameters of the bipedal robot, a multi-objective evaluation framework is established that simultaneously considers kinematic accuracy and dynamic energy consumption. The optimization variable vector is defined as:

$$\boldsymbol{P}=[r_{AD},r_{DC},r_{AB},r_{BC},r_{AF},r_{FG},r_{GH},r_{HD},r_{JG},r_{HL},r_{LN},\theta_{AB},\theta_{EF}]^{T} \tag{3-7}$$

and during the optimization process, the geometric consistency constraints of the parallelogram mechanism are strictly enforced, namely $r_{\mathrm{ML}} = r_{\mathrm{JG}}$ and $r_{\mathrm{JM}} = r_{\mathrm{GL}}$. The evaluation metrics are defined as:

Peak driving torque f_1: This metric represents the maximum load demand on the driving motor and directly influences motor selection as well as the risk of system overload.:

$$f_1(\boldsymbol{P})=\max_{t\in[t_{\mathrm{start}},t_{\mathrm{end}}]}|\tau_{\mathrm{in}}(t,\boldsymbol{P})| \tag{3-8}$$

Torque fluctuation smoothness f_2: This index evaluates the stability of the driving process through the standard deviation of the torque profile. A smaller value of f_2 indicates smoother torque variation, which helps reduce mechanical vibration and prolong the service life of the transmission components.

$$f_2(\boldsymbol{P})=\sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(\tau_{in,k}-\bar{\tau}_{in}\right)^2} \tag{3-9}$$

Average absolute power consumption f_3: This metric serves as an indicator of the system's energy consumption, representing the average power required for the robot to complete a single motion cycle.

$$f_3(\boldsymbol{P}) = \frac{1}{N}\sum_{k=1}^{N}\left|\tau_{in,k}\right| \tag{3-10}$$

Trajectory approximation error f_4: This metric functions as a hard constraint in the optimization process, ensuring that the optimized linkage parameters preserve the motion accuracy of the foot along the predefined trajectory. In this paper, the allowable error threshold is set to 10 mm:

$$f_4(\boldsymbol{P}) = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left[\left(x_N - x_{\text{ref}}\right)^2 + \left(y_N - y_{\text{ref}}\right)^2\right]} \tag{3-11}$$

In summary, the objective of the optimization process is to determine an optimal parameter vector P^* subject to the constraint $f_4(P) \leqslant 10$ mm, such that the overall performance metrics composed of $\{f_1, f_2, f_3\}$ are simultaneously improved.

3.4 Composite Decision Model Based on Improved Entropy Weighting TOPSIS

To systematically identify the optimal design parameters within the multi-objective solution space, a composite decision framework is established that integrates Differential Evolution (DE) with an improved Entropy Weighting Method (EWM) and the TOPSIS evaluation approach. The overall process is implemented in three stages:

1) Constraint-based population pre-screening: The DE algorithm is first applied to perform a global search in the parameter space, generating a candidate population $X = \{P_1, P_2, ..., P_{Np}\}$. Only the feasible solutions that satisfy the constraint $f_4(P) \leq 10$ mm are retained. These feasible candidates form the decision matrix $D_{m\times3} = [f_{ij}]$, where m represents the number of feasible solutions and $j \in \{1, 2, 3\}$ corresponds to the three evaluation metrics.

2) Improved entropy weighting for objective weighting: Because the evaluation metrics have different physical dimensions, the decision matrix is first normalized. To mitigate fluctuations in the entropy calculation when the trajectory error approaches zero (commonly referred to as the "zero-error trap"), a smoothing factor ε is introduced prior to computing the entropy value:

$$p_{ij} = \frac{f_{ij} + \varepsilon}{\sum_{i=1}^{m}\left(f_{ij} + \varepsilon\right)} \tag{3-12}$$

The weight ω_j of each metric is then determined based on its information entropy e_j:

$$e_j = -\frac{1}{\ln m}\sum_{i=1}^{m} p_{ij}\ln p_{ij}, \quad \omega_j = \frac{1 - e_j}{\sum(1 - e_j)} \tag{3-13}$$

This approach enables automatic weight allocation among the objectives, thereby improving the objectivity of the decision-making process.

3) TOPSIS relative closeness evaluation: After obtaining the weighted normalized matrix, the positive ideal solution f^+ and the negative ideal solution f^- are defined. The Euclidean distances d_i^+ and d_i^- between each candidate solution and the two ideal solutions are subsequently calculated. The candidate solutions are then ranked according to their relative closeness coefficient C_i:

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{3-14}$$

The candidate solution with the maximum C_i value—indicating the closest proximity to the ideal solution—is selected as the final optimal set of reconstructed mechanism parameters.

3.5 Optimization Results Analysis and ADAMS Benchmarking

Following the optimization flowchart illustrated in **Fig. 7**, the multi-objective optimization procedure was carried out using 50 generations of DE iterations. Within the feasible solution space that satisfies the prescribed constraints, the entropy weighting method was applied to determine the relatively important weight of each evaluation metric. The resulting objective weight distribution for the performance metrics $\{f_1, f_2, f_3\}$ is as follows: peak torque (f_1) accounts for 42%, smoothness (f_2) accounts for 35%, and energy consumption evaluation (f_3) accounts for 23%. Based on the TOPSIS relative closeness coefficient C_i, the candidate solution with the highest score was selected as the optimal reconstructed parameter set. A comprehensive analysis was then conducted between this optimized configuration and the initial reference parameter set.

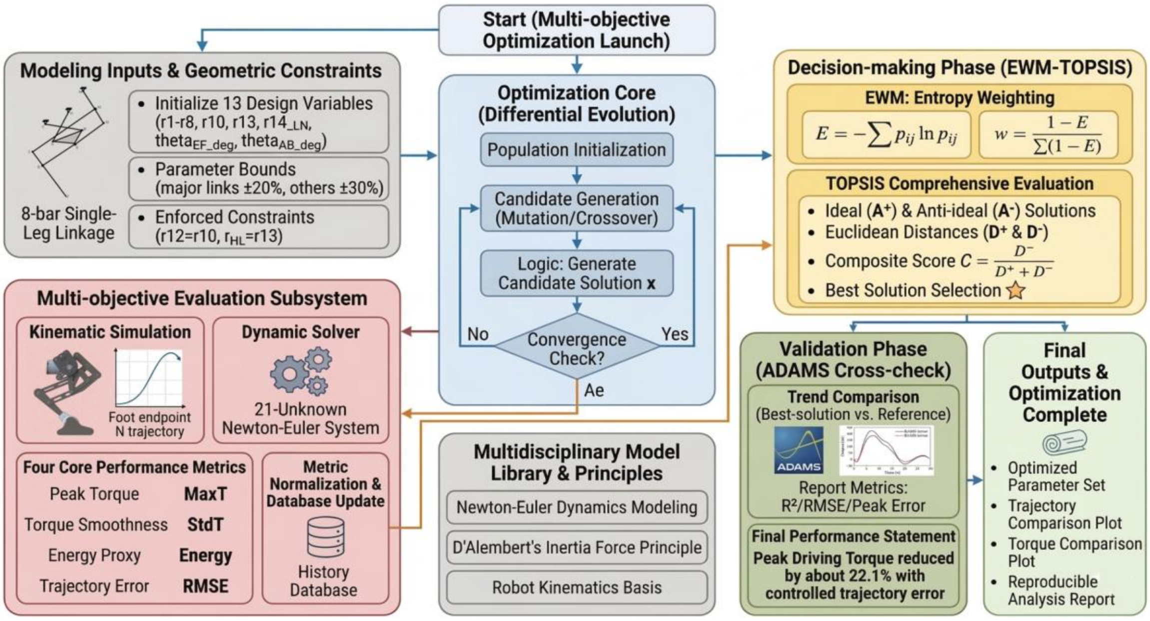


Fig. 7. Complete flowchart of the multi-objective optimization process.

3.5.1 Differential Evolution Algorithm Parameter Configuration

To ensure both convergence and global search capability during the optimization process, the DE algorithm is employed to optimize the 13 design variables of the mechanism. The key algorithm parameters are summarized in **Table 1**.

Table 1. Parameter configuration of the differential evolution algorithm

Parameters	Value	Description
Optimization strategy	best1bin	Mutation strategy based on the current best solution
Population size	15	Number of individuals per generation
Maximum iterations	50	Evolution termination condition
Mutation factor	(0.5, 1)	Scaling range of the differential vector
Recombination probability	0.7	Generation probability of the trial vector
Tolerance	0.01	Threshold for convergence determination
Parallel computing	1	Serial evaluation (dynamics solving)

During the optimization procedure, the search bounds for the primary linkage lengths are defined within ±20% of their initial reference values, while those for the remaining parameters are allowed to vary within ±30%. In addition, to preserve the geometric constraints of the parallelogram linkage MJGL, $r_{ML} = r_{JG}$ and $r_{JM} = r_{HL}$ are strictly enforced throughout the optimization process.

3.5.2 Comparison Before and After Parameter Optimization

Table 2 presents the changes in the full set of geometric parameters of the mechanism before and after optimization. The results indicate that by adjusting the length of the primary link r_{HL} and reconfiguring the installation offset angle θ_{AB}, the force transmission characteristics of the mechanism along the target trajectory are significantly improved.

Table 2. Comparison of reconstructed parameters before and after optimization.

Parameter symbol	Initial	Optimized
r_{AD} / mm	110.01	112.52

r_{DC} / mm	99.99	87.11
r_{AB} / mm	60.00	50.48
r_{BC} / mm	94.55	75.71
r_{AF} / mm	60.00	54.86
r_{JG} / mm	60.00	73.19
r_{GH} / mm	46.47	53.32
r_{HD} / mm	86.601	98.48
r_{HL} / mm	254.41	322.60
r_{LN} / mm	199.98	173.12
r_{FG} / mm	121.94	101.39
θ_{AB} / °	228.61	204.37
θ_{EF} / °	55.61	50.87

3.5.3 Dynamic Performance Analysis

The optimized mechanism demonstrates superior performance compared with the initial design across all evaluation met777rics. As summarized in **Table 3**, the peak driving torque is reduced from 1.7312 N·m to 1.3482 N·m, corresponding to a reduction of 22.1%.

Table 3. Evaluation results of dynamic optimization metrics

Parameter symbol	Initial	Optimized	Improvement percentage
Peak torque f_1 (N·m)	1.7312	1.3482	22.1%
Torque standard deviation f_2	0.4821	0.3615	25.0%
Average absolute torque f_3	0.9245	0.7431	19.6%
Trajectory RMSE f_4 (mm)	1.25	8.42	(Satisfies < 10 mm constraint)

To further validate the accuracy of the Newton-Euler theoretical model, the optimized parameters were imported into the ADAMS multibody dynamics simulation platform for comparative verification. As illustrated in **Fig. 8,** the torque profiles obtained from the analytical model and the ADAMS simulation show strong agreement throughout the entire motion cycle. The torque amplitude discrepancy remains within 3.5%, thereby confirming the validity and reliability of the 21-variable dynamic modelling approach proposed in this paper.

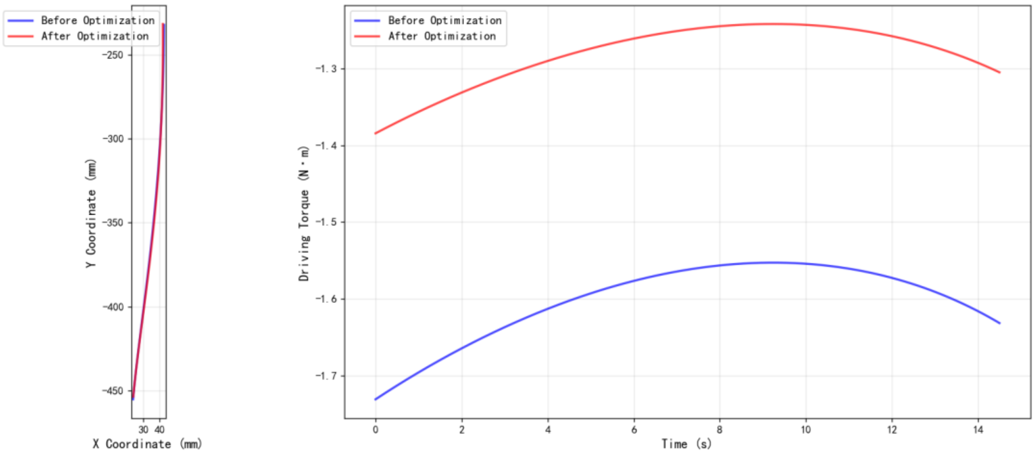


Fig. 8. Comparison of foot endpoint trajectories and driving torques before and after optimization.

4 Strategy Training and Simulation Verification

Following the completion of the mechanism parameter optimization based on the Newton-Euler dynamic model and the EWM-TOPSIS, the optimized structural parameters are imported into the simulation environment to develop a corresponding RL-based motion control strategy. As illustrated in **Fig. 9**, the overall training pipeline of the proposed RL control framework is presented.

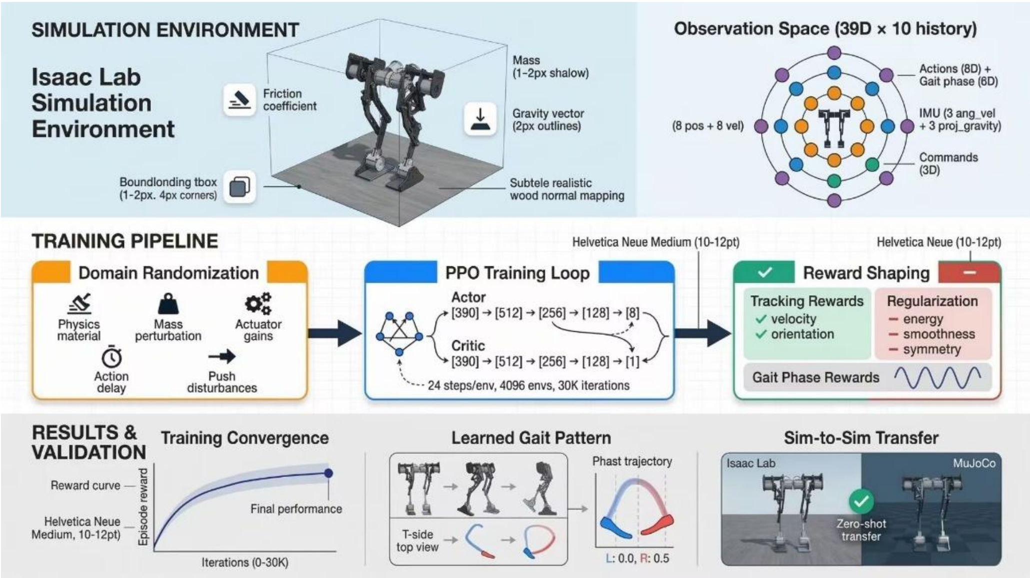


Fig. 9. Complete flowchart of the multi-objective optimization process.

4.1 Simulation Physical Environment Configuration

To develop a motion controller that can be effectively transferred to the physical prototype, a paper digital twin environment of the bipedal robot is constructed using Isaac Sim, a physics-based robotics simulation platform developed by NVIDIA.

4.1.1 Robot Model Import and Entity Calibration

The simulation model is constructed based on the optimal geometric parameter vector obtained in Section 2.5. The robot adopts a symmetrical configuration, where every single leg possesses four active degrees of freedom (DOFs): hip_roll, hip_pitch, knee_pitch, and ankle_pitch, resulting in a total of 8 DOFs for the entire. The mechanical model is first exported in URDF format and subsequently converted into the USD format required by the simulation engine. During the conversion of the closed-chain parallel mechanism, key physical parameters—including joint limits, friction coefficients, and transmission ratios—are carefully calibrated according to the actual hardware specification of the prototype.

4.1.2 Simulation Environment Parameter Configuration

The training efficiency of reinforcement learning algorithms is strongly influenced by the scale and diversity of collected samples. To improve the robustness of the learned control policy and enhance its generalization capability in real-world environments, domain randomization techniques are incorporated during the simulation training process. In each training episode, key factors such as physical parameters, initial conditions, and external disturbances are randomly perturbed. This approach enables the agent to experience a wide range of environment variations, thereby improving its resilience to modeling inaccuracies and environmental uncertainties. **Table 4** summarizes the domain randomization parameters adopted in this paper.

Table 4. Domain randomization parameter configurations

Randomization Parameter	Randomization Range	Trigger Mode
Static friction coefficient	0.6-1.0	At initialization
Dynamic friction coefficient	0.4-0.8	At initialization
Restitution coefficient	0-0.15	At initialization
Base mass perturbation	0.8-2.0 kg	On every reset
Initial linear velocity v_x	0.1-0.2 m/s	On every reset
Joint initial position scale	1.0	On every reset
External push F_x, F_y	-0.5-0.5 m/s	10-15s interval

The configuration of the physical parameters in the simulation environment plays a crucial role in determining both the realism of the simulation and the transferability of the trained policy to the physical robot. **Table 5** presents the key simulation parameters used in this paper along with the rationale for their selection.

Table 5. Isaac Lab simulation environment physical parameter settings

Parameter Name	Value
Parallel environments (num_envs)	4096
Physics simulation frequency (sim_dt)	200
Control frequency (decimation)	50
Ground friction coefficient (friction)	0.8
Maximum episode duration (max_episode)	20.0
Physics engine	PhysX

High-frequency contact sensors are developed at the foot endpoints to monitor ground contact states in real time. This large-scale sampling mechanism enables the agent to experience a substantial number of falling and recovery events within a short period, thereby significantly accelerating the convergence of the control policy.

4.2 Reinforcement Learning Algorithm and State-Action Space Definition

To address the challenges of dynamic balance and motion planning for the bipedal robot, the Proximal Policy Optimization (PPO) algorithm is adopted to establish the reinforcement learning control framework. By introducing a clipped objective function that restricts the policy update step size, PPO ensures stable policy evolution during gait learning while maintaining high sampling efficiency in large-scale parallel training environments.

4.2.1 Observation Space and Historical Frame Mechanism

To enable the agent to perceive the dynamic characteristics of the mechanism and incorporate temporal memory, a high-dimensional observation space O_t is constructed. The observation vector for a single frame contains 39 feature dimensions. To compensate for sensor transmission delays and to approximate the Markov property of the system, an observation history buffer with a length of H=10 is introduced. The final input observation vector $\phi_t \in \mathbb{R}^{390}$ is therefore defined as:

$$\phi_t = \left[\boldsymbol{o}_t, \boldsymbol{o}_{t-1}, \ldots, \boldsymbol{o}_{t-H+1}\right]^T \tag{4-1}$$

The detailed dimensional allocation of the single-frame observation feature o_t is summarized in **Table 6**:

Table 6. Single-frame observation feature dimension allocation

Observation Feature	Dimension	Physical Meaning
Angular velocity ω_b	3	Angular velocity in the body coordinate system
Projected gravity g_b	3	Gravity vector in body frame
Velocity command v_{cmd}	3	Desired v_x, v_y, and yaw angular velocity ω_z
Joint position q	8	Current angular positions of the 8 active joints
Joint velocity v	8	Angular rate of change of all active joints
Previous action a_{t-1}	8	Previous position commands
Gait phase information	6	Sin, cos, and cycle ratio of the left and right leg phases
Total	39	Total observation dimension = 39 x 10 = 390

4.2.2 Action Space and Control Law

The action space $\mathbf{A} \in \mathrm{R}^8$ directly corresponds to the 8 actively driven joints of the robot. To comply with the safe operating limits of the hardware actuators and to improve the smoothness of the learning process, the neural network output $a \in [-1,1]$ is not applied directly as torque. Instead, it is scaled by an action scaling factor $\sigma_{act} = 0.25$ to produce the desired joint angular offset $\Delta\mathbf{q}_{target}$. The low-level actuator control follows a PD control law:

$$\boldsymbol{\tau} = K_p\left(\boldsymbol{q}_{\text{default}} + \sigma_{\text{act}} \cdot \boldsymbol{a} - \boldsymbol{q}\right) - K_d\,\dot{\boldsymbol{q}} \tag{4-2}$$

where q_{default} denotes the nominal initial posture defined in Section 1. This strategy allows the robot to maintain a stable standing configuration during the early stages of training, while progressively exploring stable gait deviations as the training process advances.

4.3 Reward Function Design and Simulation Performance Analysis

To guide the agent toward learning an efficient and stable walking gait while satisfying the closed-chain dynamic constraints of the mechanism, a composite reward function is constructed. The reward formulation integrates task tracking objectives, physical constraint penalties, and gait guidance $R = \Sigma w_i r_i$ to jointly shape the learning process.

4.3.1 Definition and Weight Allocation of the Composite Control Reward Function

The key reward components and their corresponding weight allocations adopted in this paper are summarized in **Table 7**:

Table 7. Reinforcement learning reward function configuration

Reward Term(r_i)	Weight(w_i)	Calculation Method	Optimized Physical Objective
Tracking Lin Vel	6.0	$exp\left(-\lvert\Delta v\rvert^2 / \sigma^2\right)$	Maintain steady forward velocity
Tracking Ang Vel	0.3	$\exp\left(-\lvert\boldsymbol{\omega}_{z,cmd} - \boldsymbol{\omega}_{z,base}\rvert^2 / \sigma_{\boldsymbol{\omega}}^2\right)$	Minimize yaw angle deviation
Energy/Torque	-1e-4	$\sum\lvert\tau \cdot \dot{q}\rvert$	Minimize energy consumption
Action Rate	-0.05	$\sum\lvert a_t - a_{t-1}\rvert^2$	Reduce joint jitter & smooth motion
Joint Accel	-1e-6	$\sum\lvert\ddot{q}\rvert^2$	Protect the mechanical lifespan of the mechanism
Gait Cycle	2.0	$\cos(\phi_L - \phi_R - \pi)$	Guide the generation of a human-like gait cycle
Hip Deviation	-0.25	$\sum\lvert q_{hip}\rvert^2$	Reduce lateral CoM deviation
Base Height	-1.0	$\lvert h - h_{ref}\rvert^2$	Prevent excessive squatting or jumping actions

1. Primary task tracking reward: This term incentivizes the agent to accurately follow the commanded velocity objectives. In particular, the linear velocity tracking reward is formulated using an exponential function to ensure balanced task incentives across different target speeds:

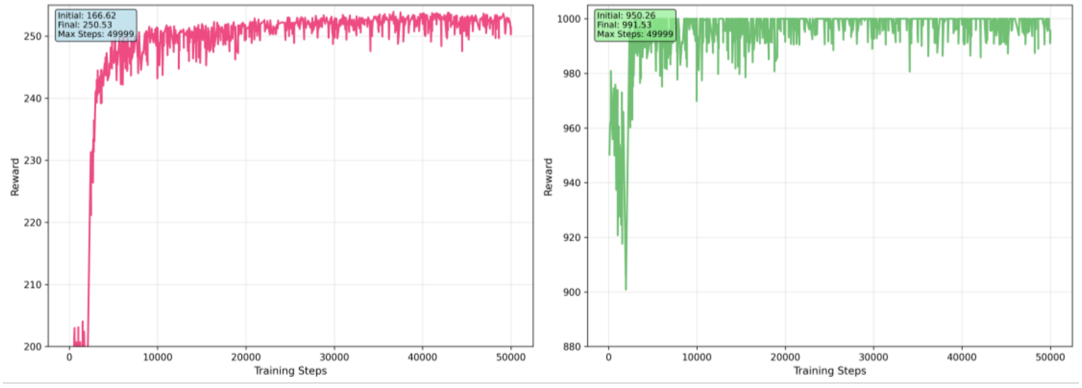
$$r_{\text{lin}_{\text{vel}}} = w_{\text{lin}} \exp\left(-\frac{\|\boldsymbol{V}_{xy,\text{cmd}} - \boldsymbol{V}_{xy,\text{base}}\|^2}{\sigma_{\text{vel}}^2}\right) \tag{4-3}$$

2. Dynamic energy efficiency and smoothness reward: This term is directly aligned with the dynamic optimization objectives described in Section 3, aiming to improve energy efficiency by penalizing the instantaneous power consumption of the motors as well as excessive joint oscillations:

$$r_{\text{energy}} = w_e \cdot \sum_{j=1}^{8} \|\tau_j \cdot \dot{q}_j\| \tag{4-4}$$

where the penalties imposed on the action rate and joint acceleration serve to enforce smooth torque outputs and suppress abrupt control variations.

3. Gait cycle and symmetry constraints: Considering the 8-joint symmetric structure of the robot, a gait phase guidance term r_{gait} is incorporated into the reward function. In addition, a hip deviation penalty $r_{\text{hip\_dev}}$, is introduced to limit excessive lateral body motion. This constraint enhances dynamic stability during locomotion. Under the guidance of the above composite reward formulation, the control policy is trained through large-scale parallel iterative learning within the Isaac Sim environment. The convergence behavior of the total reward during the training process is illustrated in **Fig. 10**.



(a) (b)

Fig. 10. Training curves in Isaac Sim: (a) reward value vs. training iterations, (b) episode length (steps) vs. training iterations.

4.3.2 Evaluation of Simulated Gait Performance

After 50,000 training iterations conducted across 4,096 parallel environments, the control policy learned by the agent demonstrated robustness within the Isaac Lab. As illustrated in **Fig. 11**, sequential simulation snapshots under a preset command velocity (v_x = 0.3m/s) show that the prototype robot is capable of maintaining a stable, human-like gait. Throughout the simulation, the amplitude of the lateral body sway remains within a small range, visually confirming the effectiveness of the integrated design approach that combines the optimized mechanical configuration with the learned control policy.

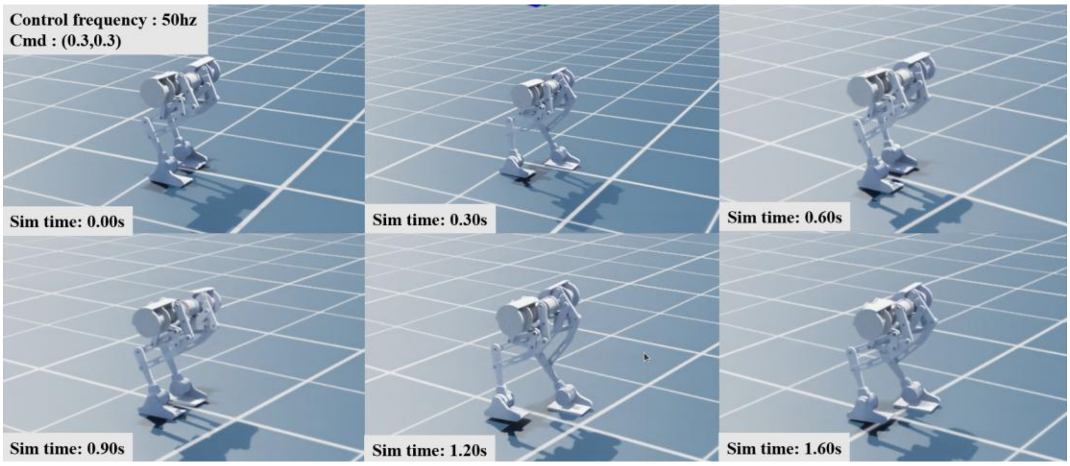


Fig. 11. Validation of the training policy in the Isaac Sim environment.

4.4 MuJoCo-based Sim-to-Sim Cross-Engine Consistency Validation

To assess the generalization capability of the trained control policy in environments outside the training platform and to mitigate potential biases associated with the dynamic solution methods of a single simulation engine—particularly for closed-chain parallel mechanisms—a Sim-to-Sim validation framework based on MuJoCo is established.

4.4.1 Cross-Engine Model Migration and Calibration

The high-dimensional control policy trained in Isaac Lab is exported as standardized neural network weight parameters. Within the MuJoCo simulation environment, the robot model is reconstructed using the same optimized parameter set P*. MuJoCo employs a different generalized-coordinate multibody dynamics solver, which offers more refined modeling of contact interactions and joint friction characteristics.

4.4.2 Sim-to-Sim Validation Results

In the MuJoCo simulator, the exported controller is directly deployed on the reconstructed robot model under a commanded forward velocity of v_x = 0.3 m/s. The experimental results demonstrate the following:

Policy transfer success rate: As illustrated in **Fig. 12**, the controller achieves long-distance stable walking in MuJoCo without any additional fine-tuning, representing a successful zero-shot transfer. This result confirms the strong cross-platform robustness of the learned control policy.

Gait behavior consistency: Although MuJoCo and Isaac Lab employ fundamentally different contact models and physics solvers, the robot maintains a human-like alternating gait pattern and stable body posture. The consistency in macro-level kinematic performance indicates that the proposed closed-chain mechanism demonstrates strong adaptability and numerical stability across different physics engines.

Disturbance rejection: When a lateral impulse disturbance of 10 N·s is applied to the robot body in MuJoCo, the control policy enables rapid posture recovery and dynamic balance adjustment. The observed behavior aligns with the expected disturbance-rejection performance of the physical prototype.

This two-stage Sim-to-Sim validation framework effectively mitigates the “simulation island” effect that may arise from reliance on a single physics engine, thereby providing a deterministic foundation for the physical prototype experiments presented in Section 5.

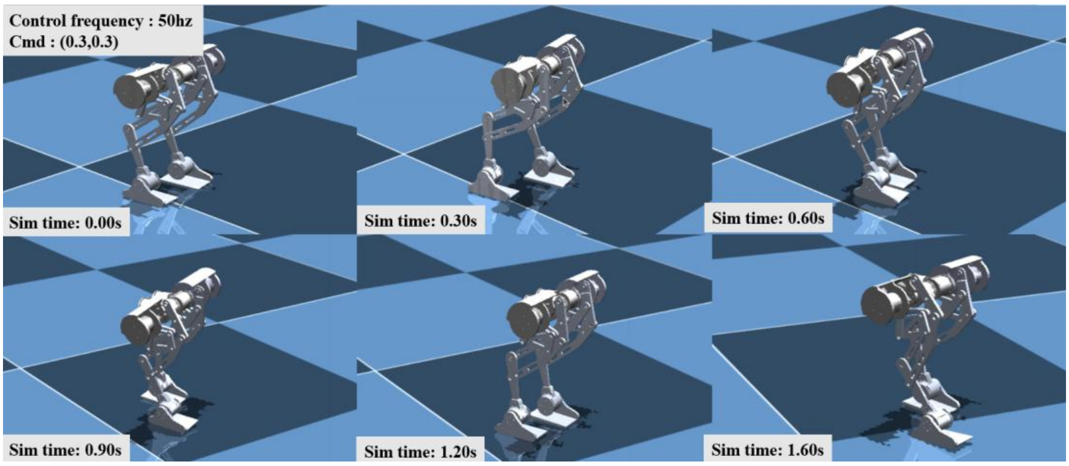


Fig. 12. Validation of the training policy in the MuJoCo environment.

5 Prototype Performance Testing and Simulation Consistency Verification

To validate the optimization model and control strategy proposed in this paper, an experimental prototype of an 8-DOF closed-chain parallel bipedal robot was developed, and a series of physical experiments were conducted for verification.

5.1 Hardware System Implementation

5.1.1 Prototype Design

The physical prototype adopts a bipedal configuration with four active DOFs per leg. From the proximal to distal joints, these include the hip_roll, hip_pitch, knee_pitch, and ankle_pitch joints, resulting in a total of 8 active DOFs for the entire robot. Among these, the knee joint incorporates the single-DOF closed-chain 8-bar transmission mechanism proposed earlier in this work. The geometric dimensions of the links strictly follow the optimal parameter configuration obtained in Section 3.5. In addition, structural anti-interference optimization was carried out during the mechanical design process to ensure reliable operation during typical walking motions. Accordingly, the effective physical range of motion for each active joint is reasonably constrained to approximately ±30°, which satisfies the fundamental kinematic requirements for stable bipedal locomotion.

To ensure sufficient structural rigidity while maintaining a lightweight design, the primary links of the prototype are fabricated from aluminum alloy using CNC machining. The connectors interfaces between the joint motors and the links employ a hybrid structure composed of carbon-fiber plates and 3D-printed nylon components. This combination effectively reduces the overall mass of the system while preserving adequate structural stiffness. In terms of actuation, each active joint is equipped with a compact integrated servo joint motor. All remaining passive joints within the closed-chain mechanism are supported by precision deep-groove ball bearings, which minimize the influence of mechanical friction on the system's dynamic behavior. To provide a comprehensive overview of the hardware platform, **Table 8** summarizes the key configuration parameters of the experimental prototype, including major specifications such as mechanical dimensions, control units, drive modules, and sensing devices.

Table 8. Key hardware configuration of the experimental prototype

Parameter	Value	Parameter	Value
Weight (kg)	7.6	Max standing height (m)	0.45
High-level computer	NUC	CPU	Intel® Core™ Ultra 5
Low-level computer	DM-MC02	Main control chip	STM32H723VGT6
Motor model	DM-4340P	Max torque (N·m)	28
Motor stiffness KP	25	Motor damping KD	1.5
Max input voltage (V)	24	Max current limit (A)	8
Power supply type	Lithium battery	Capacity (mAh)	3300
MU model	DM-IMU-L1	Frequency (MHz)	200

5.1.2 Control System Design

As illustrated in the hierarchical control architecture shown in **Fig. 13**, the prototype adopts a dual-layer control framework consisting of a high-level and a low-level controller. The high-level controller operates within an Ubuntu 20.04 + ROS1 Noetic environment. It processes high-level sensory inputs and performs real-time inference using the PPO neural network policy trained in Section 4 to generate motion decisions. The trained policy model is deployed using the ONNX Runtime inference engine on the high-level computing unit, which receives observation data and outputs action commands at a control frequency of 100 Hz. The core software architecture includes several functional modules: the RL controller, responsible for executing reinforcement learning inference; the PD controller, which implements the underlying fast control loop; the state machine, used to manage the robot's operational modes; and the IMU driver, which acquires attitude and motion data from the inertial sensor.

The low-level control system employs the DM-MC02 development board as the onboard real-time embedded control unit. Powered by a high-performance STM32H7 processor, the DM-MC02 operates under the FreeRTOS real-time operating system and is responsible for executing the joint target commands issued by the high-level controller. Communication between the high-level and low-level systems is implemented through a serial interface. The high-level controller transmits action commands at a frequency of 100 Hz, while the low-level controller continuously returns joint state information and IMU measurements in real time. To enhance communication reliability and real-time performance, the low-level system adopts a dual-channel FDCAN communication architecture. Specifically, the four DM4340P integrated servo joint motors on the left leg are controlled via one FDCAN bus, while the four motors on the right leg are driven through a separate FDCAN bus. This design effectively reduces bus load and increases control bandwidth. At the bottom layer, the control framework implements FOC-based closed-loop regulation of current, velocity, and position. In addition, the low-level controller integrates a BMI088 IMU sensor to obtain real-time body attitude data. Motor rotational position and velocity are measured through magnetic encoders and transmitted to the low-level controller for processing. These measurements are subsequently relayed to the high-level controller via the serial communication channel, forming a complete closed-loop control architecture for the bipedal robot. This hierarchical system architecture effectively balances the computational demands of neural network inference with the strict real-time requirements of the underlying drive.

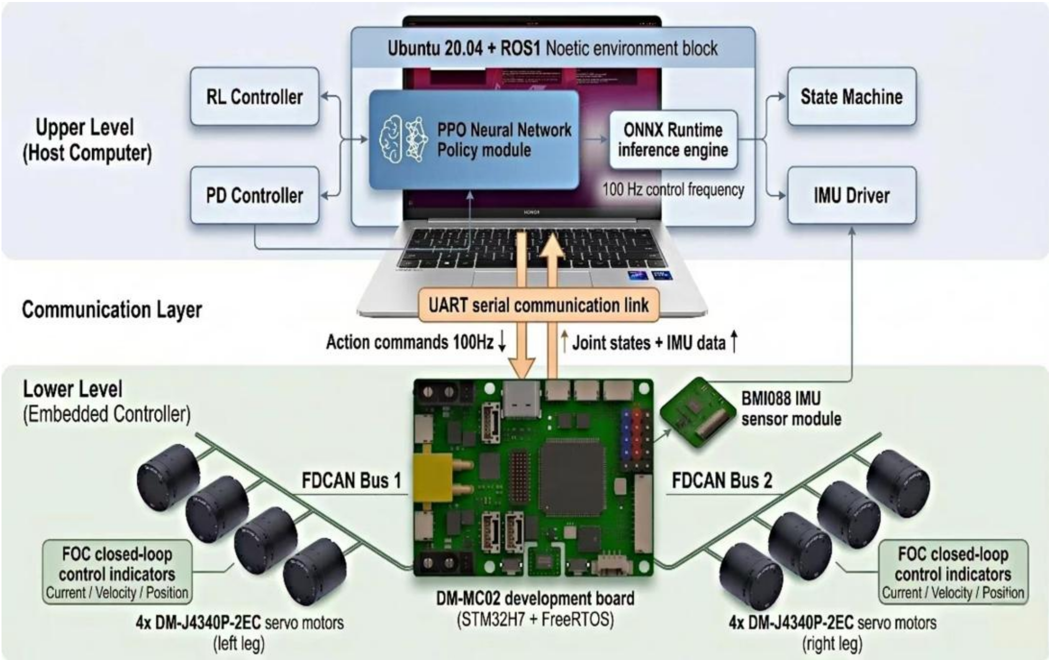


Fig. 13. Architecture diagram of the hierarchical control system.

5.1.3 Robot Prototype

As illustrated in **Fig. 14**, the fully assembled robot prototype has a total mass of 7.6 kg. The mass distribution and moments of inertia of each link were experimentally calibrated and subsequently incorporated into the simulation environment. This calibration ensures a high level of physical consistency between the digital twin model and the physical prototype.

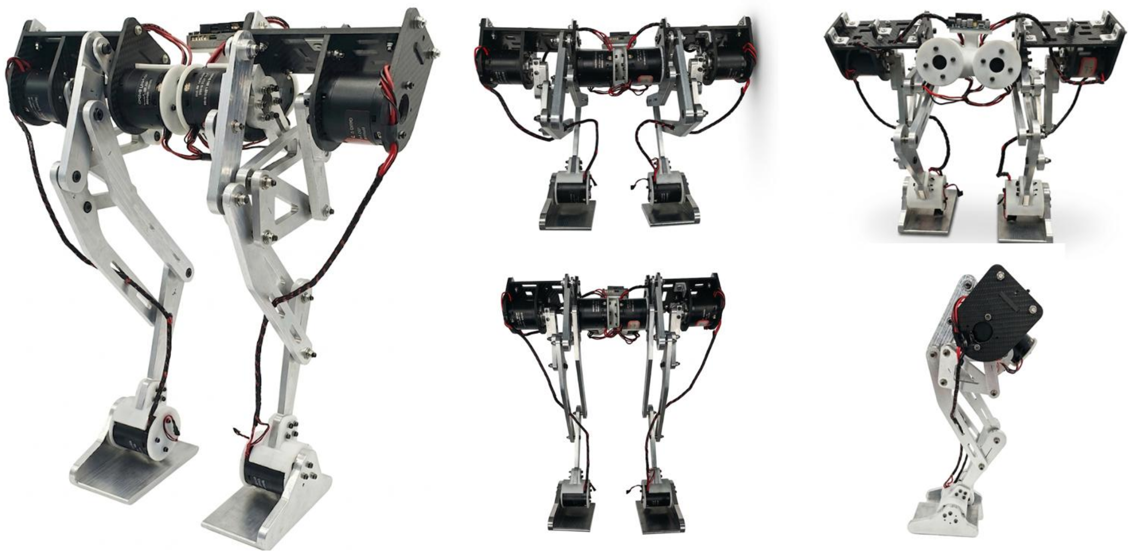


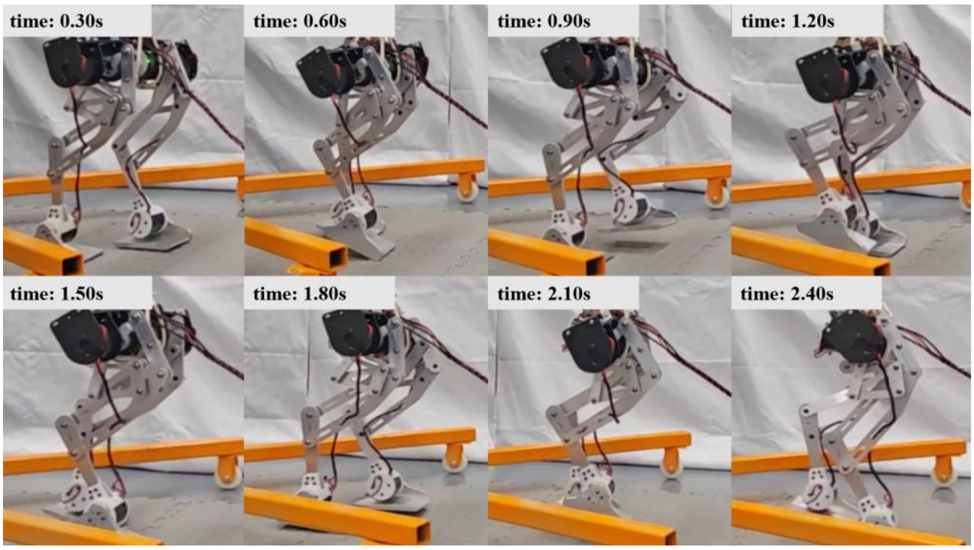
Fig. 14. Physical prototype.

5.2 Experimental Scenario Design and Simulation Benchmarking Analysis

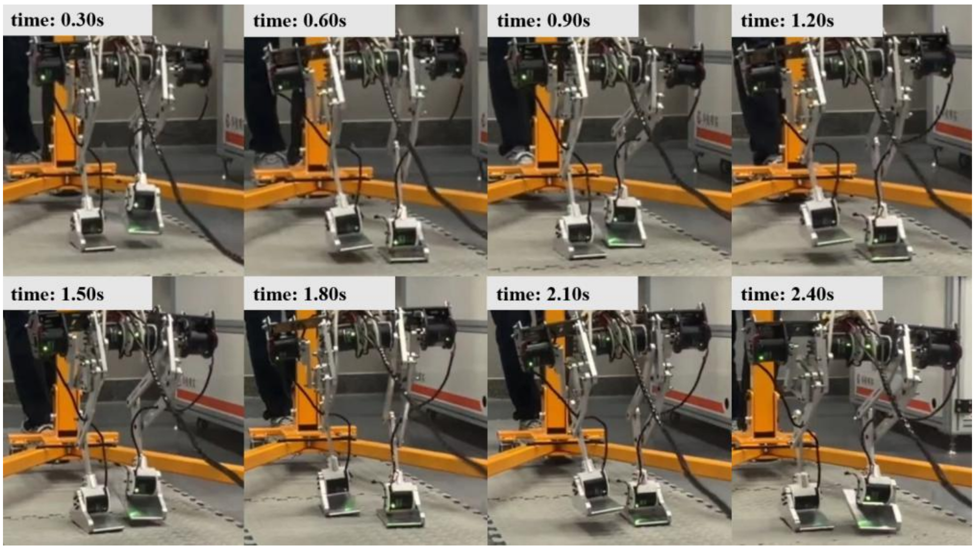
In this section, the motor performance of the optimized prototype is evaluated through physical experiments conducted on flat laboratory ground. The experimental results are then benchmarked against the simulation predictions obtained from the Isaac Lab.

5.2.1 Empirical Analysis of Motion Stability

During the experiments, the onboard IMU continuously recorded the attitude variations of the prototype while executing a commanded forward velocity of v_x = 0.3 m/s. As shown in the body IMU attitude and acceleration curves in **Fig. 16(a)**, the experimental results indicate that the peak fluctuation ranges of the body around the horizontal roll axis (Roll) and pitch axis (Pitch) remain within ±2.5° and ±4.0°, respectively. These magnitudes are comparable to the natural trunk motion observed in human walking. In addition, the Sim-to-Real continuous gait snapshots presented in **Fig. 15** visually illustrate the full sequence of the robot's smooth walking motion. The observed periodic and stable attitude variations demonstrate that the reinforcement learning control policy, when deployed directly to the physical prototype without additional fine-tuning, effectively compensates for structural interference and assembly imperfections. As a result, the single-leg closed-chain bipedal robot maintains reliable dynamic balance during locomotion.



(a)



(b)

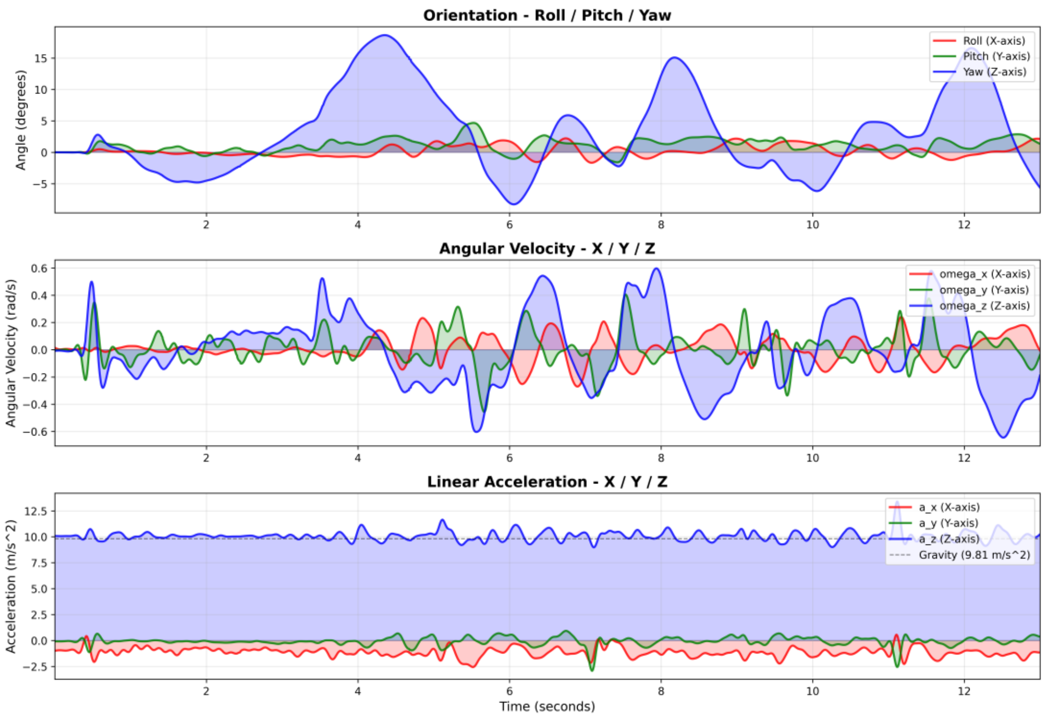
Fig. 15 Continuous gait snapshots of the robot during walking. (a) side view, (b) front view.

5.2.2 Empirical Validation of Driving Performance and Energy Efficiency

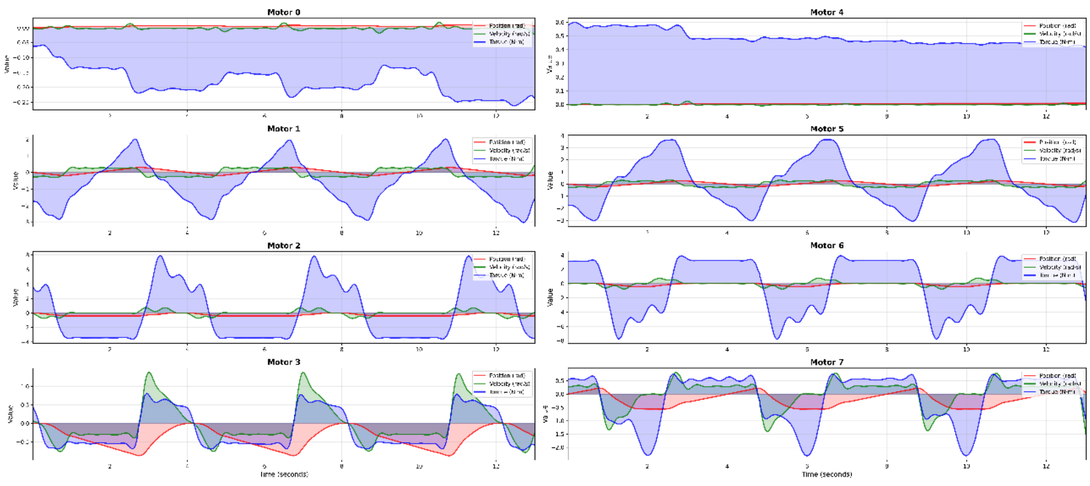
Based on the motor state feedback data of the left and right legs shown in **Figs. 16(b) and (c)**, the actual joint torque and velocity profiles of the robot over a standard gait cycle were extracted and analyzed. The experimental results indicate that, due to practical factors such as assembly tolerances of the physical prototype, unmodeled joint friction, and delays in the low-level control loop, certain discrepancies (Sim-to-Real gap) exist between the measured torque curves and the ideal rigid-body Newton-Euler dynamic predictions presented in Section 3.

Nevertheless, the measured torque profiles preserve clear periodic characteristics that correspond closely to the phases of the gait cycle. More importantly, during continuous gait transitions and phases of unilateral load-bearing support, the torque outputs of all joint motors remain within a stable and safe operating range. No torque saturation or high-frequency oscillations were observed throughout the experiments. The smooth and well-bounded actuator outputs not only demonstrate the strong robustness of the reinforcement learning control policy when confronted with unmodeled physical disturbances, but also indirectly confirm that the proposed closed-chain mechanism optimization effectively reduces extreme peak loads on the motors. As a result, the prototype maintains high force transmission efficiency and operational safety during real-world locomotion.

The strong agreement between the experimental measurements and the simulation predictions validates the accuracy of the proposed dynamic model. This consistency indirectly supports the theoretical results obtained during the simulation stage, indicating that the optimized configuration achieves a 22.1% reduction in peak torque compared with the initial design. Consequently, these findings further confirm that the link proportion scheme derived from 21-axis dynamic optimization exhibits clear structural advantages during practical physical operation.



(a)



(b) (c)

Fig. 16. Experimental results of locomotion stability and actuator performance: (a) Body IMU attitude and acceleration, (b) left leg motor state feedback, and (c) right leg motor state feedback.

6 Conclusions

This paper presents a systematic framework for the integrated design and control of closed-chain bipedal robots.A parameter optimization framework is proposed that combines a 21-unknown global Newton-Euler dynamic matrix with an entropy-weighted TOPSIS decision-making approach. The results demonstrate that optimizing the linkage geometry can fundamentally improve force transmission characteristics, with the proposed design-stage optimization alone reducing peak driving torque by 22.1%. In addition, a model-free reinforcement learning control strategy was developed through large-scale simulation training in Isaac Lab, followed by rigorous Sim-to-Sim cross-engine validation using MuJoCo. The optimized design and trained policy were subsequently implemented on a physical prototype, which successfully achieved stable dynamic walking and reliable Sim-to-Real policy transfer. The experimental results are generally consistent with the simulation predictions, supporting the validity of the proposed dynamic modeling and optimization framework. These findings provide preliminary evidence that the presented methodology offers an effective solution for the integrated mechanism design and locomotion control of highly dynamic legged robotic systems.

Nomenclature

Variables

M	Total degrees of freedom
d	Dimension of motion space
n	Number of moving components (links)
g	Number of joints
v	Number of redundant geometric constraints
r_{AD}	Length of linkage AD (input crank)
r_{DC}	Length of linkage DC
r_{AB}	Length of fixed linkage AB
r_{BC}	Length of linkage BC
r_{AF}	Length of linkage AF
r_{FG}	Length of linkage FG
r_{GH}	Length of linkage GH
r_{HD}	Length of linkage HD
r_{IH}	Length of linkage IH
r_{JG}	Length of linkage JG (parallelogram member)
r_{ML}	Length of linkage ML (parallelogram constraint)
r_{HL}	Length of linkage HL
r_{LN}	Length of linkage LN (foot extension)
θ_{AD}	Input angle of driving joint A (crank angle)
θ_{DC}	Angular position of linkage DC
θ_{BC}	Angular position of linkage BC
θ_{EF}	Angle of linkage EF
θ_{FG}	Angle of linkage FG
θ_{GH}	Angle of linkage GH
θ_{IH}	Angle of linkage IH
θ_{JG}	Angle of linkage JG
θ_{ML}	Angle of linkage ML
θ_{JM}	Angle of linkage JM
θ_{GL}	Angle of linkage GL
θ_{AB}	Installation offset angle
φ_{bias}	Bias angle for coupling between loops
θ_{hip_roll}	Hip_roll joint angle
θ_{hip_pitch}	Hip_pitch joint angle
θ_{knee}	Knee_pitch joint angle
θ_{ankle}	Ankle Pitch joint angle
$P_{N,loca}$	Local 2D position vector of foot end N in the mechanism plane

$P_{\mathrm{N,3D}}$	3D position vector of foot end N prior to Hip_pitch rotation
P_{foot}	Global 3D position vector of foot end N in the base coordinate frame
P_{N}	Position vector of the foot end point N relative to the origin
P_{F}	Position vector of joint F relative to the origin
x_{N}, y_{N}, z_{N}	Coordinates of foot end point N
x_{G}, y_{G}	Coordinates of component center of mass
$P_{\mathrm{Hip\_pitch}}$	Position relative to Hip_pitch joint
$P_{\mathrm{Hip\_roll}}$	Position relative to Hip_roll joint
$R(\theta)$	2D rotation matrix by angle theta
R_x, R_y	3D rotation matrices about X and Y axes
m_i	Mass of the i-th linkage component
I_{tri}	Rotational inertia of a triangular rigid body component
I_i	Moment of inertia of component i
$F_{\mathrm{A}x}$, $F_{\mathrm{A}y}$	Force components at joint A
F_{ext}	External forces
$m_i g$	Gravitational force on component i
τ_{in}	Driving torque (motor output torque) from input crank
$\tau_{\mathrm{in\_bar}}$	Average driving torque over cycle
w_i	Angular velocity of linkage i
a_i	Angular acceleration of linkage i
q	Joint position vector
$\dot{q}$	Joint velocities vector
$\ddot{q}$	Joint accelerations vector
a_{Gi}	Linear acceleration of center of mass of component i
$\mathrm{A}(\theta)$	Coefficient matrix from Newton-Euler equations
t_{start}	Start time of the motion cycle
t_{end}	End time of the motion cycle
x_{ref}	X-coordinate of the reference ideal trajectory
y_{ref}	Y-coordinate of the reference ideal trajectory
P	Design parameter vector
f_1	Peak driving torque objective
f_2	Torque smoothness objective
f_3	Torque smoothness objective
f_4	Trajectory approximation error objective
f_{ij}	Criterion j value for solution i
p_{ij}	Normalized probability in TOPSIS
e_j	Information entropy of criterion j
w_j	Objective weight of criterion j
d_i^+, d_i^-	Euclidean distances to positive/negative ideal solutions
C_i	Relative closeness coefficient
O_t	Complete observation space
o_t	Single-frame observation vector at time t
ϕ_t	Observation with history buffer
v_{cmd}	Velocity command
A	Action space
a	Neural network output action
$\Delta q_{\mathrm{Target}}$	Target joint angle offset
σ_{act}	Action scaling coefficient
R	Total cumulative reward
r_i	Individual reward term i
w_i	Weight/importance coefficient for reward term i
$r_{\mathrm{lin\_vel}}$	Linear velocity tracking reward term
$\mathrm{w_{lin}}$	Weight coefficient for the linear velocity tracking reward
$\mathrm{r_{energy}}$	Energy consumption penalty term (based on joint torque and velocity)
w_e	Weight coefficient for the energy consumption penalty
r_{gait}	Gait phase symmetry guidance reward term
$r_{\mathrm{hip\_dev}}$	Hip joint deviation (lateral imbalance) penalty term

Acknowledgements The authors declare no acknowledgments.

Funding This work was supported by the National Natural Science Foundation of China (Grant No. 52575002) and the Fundamental Research Funds for the Central Universities(Grant No. 2024JBZY005)

Conflict of Interest The authors declare no conflict of interest.

Author Contributions Jianxu WU: Conceptualization, Methodology, Writing – Original Draft.
Yichen ZHANG: Conceptualization, Supervision, Writing – Review and Editing.
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Longxiang XU: Investigation.

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